

Practice Quiz: Systems

Part A: Multiple Choice

1. In a crochet project viewed as an Active Inference system, what represents the **Markov blanket**? A) The completed rows of fabric B) The yarn still in the skein C) The live working edge — hook, live loop, and working row D) The pattern instructions on paper
2. When you frog (rip back) several rows of crochet, what happens from a systems perspective? A) External states become internal states B) Internal states are converted back to external states through the blanket C) The Markov blanket becomes stronger D) Nothing changes — the system remains the same
3. The yarn tension you feel through your fingers while crocheting is best described as: A) An internal state of the fabric B) A sensory state at the Markov blanket C) An external state unrelated to the system D) A planning variable
4. Which of the following is an **internal state** of a crochet system? A) The ball of yarn in the skein B) The pattern written on paper C) Three completed rows of double crochet D) The lighting in the room
5. A gauge swatch helps the crocheter calibrate the system. In Active Inference terms, the swatch primarily helps establish: A) The color of the yarn B) The relationship between agent actions and system outcomes at the boundary C) How many projects to make D) The social dynamics of the crochet circle
6. When crocheting in the round, the Markov blanket of the working edge: A) Disappears entirely B) Forms a closed continuous loop rather than an open row C) Doubles in size D) Becomes identical to flat-row crochet
7. If the live loop slips off the hook and the fabric begins to unravel, this is analogous to: A) The system successfully minimizing free energy B) The Markov blanket dissolving, threatening system integrity C) The agent choosing a new policy D) The external states becoming more organized

Part B: Short Answer

1. Explain how the working yarn — the strand running from the skein to your hook — functions as a channel connecting external states to internal states through the Markov blanket. Use specific crochet actions to illustrate.
2. You are crocheting a scarf and you reach the end of your first ball of yarn. Describe this moment in terms of systems: what external state has changed, how does the crocheter detect it at the boundary, and what action at the blanket is required to continue the system?
3. Two crocheters share a single ball of yarn, each pulling from one end to work on separate projects. Describe the system boundaries. How many Markov blankets are there, and how do the two systems interact through shared external states?